

Jeonggyu Hwang

Graduate Student

EDUCATION

March 2025 - Present Combined Master-Doctor Student in EEE,
Yonsei University

- Supervisor: Prof. Lee, Taeyoon Ph.D.
- Research topic: TBD

RESEARCH EXPERIENCE

January 2025 - Present Graduate Research Assistant
NanoBio Device Lab. (NBDL), Yonsei University

ACADEMIC EXPERIENCE

February 2026 - June 2026 Teaching Assistant (TA)
Physical Electronics (EEE3210), Spring 2026

January 2026 - Present Administrative Assistant
EEE, Yonsei University

HONORS & AWARDS

November 2025 Priority Selection for Yonsei Graduate school

PERSONAL INFO

Address

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Seoul, Republic of Korea (03722)

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Portfolio

<https://developer-jg.github.io/>

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RESEARCH INTERESTS

- Semiconductor Device Physics
- Strongly Correlated Electron Systems
- Oxide Electronics
- Thin Film Growth and Interface Engineering

EDUCATION

Yonsei University

Combined Master-Doctor Student

School of Electrical & Electronic Engineering, College of Engineering

Seodaemun-gu, Seoul, South Korea

March 2026 - Present

Gachon University

Undergraduate Student

Major in Semiconductor Display, Department of Semiconductor Engineering,

College of Semiconductor

- Bachelor of Engineering (B.E.)
- GPA (Overall): 4.33 / 4.50 (Major GPA: 4.37 / 4.50)
- Honors: Cum Laude

Seongnam-si, Gyeonggi-do, South Korea

March 2023 - February 2026

Seoul National University

Undergraduate Exchange Student

Department of Materials Science and Engineering, College of Engineering

Gwanak-gu, Seoul, South Korea

August 2024 - February 2025

PUBLICATIONS

Conference:

- [1] **Hwang, J.**, & Jeon, J. (2025). Exploring the theoretical limits of efficiency in multilayer solar cells. Poster presented at the 32nd Korean Conference on Semiconductors (KCS), Korea.

Pre-Print:

- [3] **Hwang, J.** (2025). Estimation of Solar Spectral Irradiance Using Meteorological Data and Analysis of Optimal Conditions for Solar Power Generation. *arXiv preprint arXiv:2504.08008*.
- [2] **Hwang, J.** (2024). Theoretical analysis of doping concentration gradients on solar cell performance. *arXiv preprint arXiv:2407.12245*.
- [1] **Hwang, J.** (2024). Exploring the theoretical limits of efficiency in multilayer solar cells. *arXiv preprint arXiv:2404.14930*.

ACADEMIC EXPERIENCE

Yonsei University

Graduate Researcher

School of Electrical & Electronic Engineering, College of Engineering

- NanoBio Device Lab. (NBDL)
- PI: Prof. Lee, Taeyoon
- Research topic: TBD

Seodaemun-gu, Seoul, South Korea

January 2026 - Present

Undergraduate Researcher
Department of Physics, College of Science

June 2025 - July 2025

- Center for Bandstructure Engineering
- PI: Prof. Kim, Keunsu
- Research topic: 2D Materials and Solid-State Physics

POSTECH

Pohang-si, Gyeongsangbuk-do, South Korea

Pohang University of Science and Technology

Undergraduate Researcher

February 2024

Department of Chemical Engineering

- FIMDEM Lab. - Functional Inorganic Materials Design & Manufacturing
- PI: Prof. Son, Jaesung
- Research topic: Solution-Processed Thin Films and Thermoelectric Device Materials

Korea University

Seongbuk-gu, Seoul, South Korea

Undergraduate Researcher

July 2023 - August 2023

Department of Integrative Energy Engineering, College of Engineering

Graduate School of Energy and Environment

- Electrochemistry Lab. - Solarrino
- PI: Prof. Jun, Yongseok
- Research topic: Perovskite Solar Cells

INDUSTRY EXPERIENCE

Philoptics Co., Ltd

Osan-si, Gyeonggi-do, South Korea

Engineer

March 2024 - May 2025

Semiconductor Department, R&D Division

- Development of Advanced Glass Singulation Processing Technology
- Development of Next-Generation ABF Drilling Core Technology
- 2024 Advanced Technology Development Project

AWARDS AND HONORS

Graduation with Honors, Gachon University

Selected for Priority Admission, Yonsei University

(Waived major-specific interview and exam based on document evaluation)

Academic Excellence Scholarship, Gachon University (5 Consecutive Semesters)

Undergraduate Poster Presentation Award,

32nd Korean Conference on Semiconductor (KCS 2025)

LEADERSHIP & ACTIVITY

Research Service & Leadership

The Gachon Research Society

Honorary Member

February 2025 - Ongoing

Chairperson of the Acting Executive & Steering Committee for

the 2025 Delegates Joint Meeting

February 2025 - November 2025

Executive Adviser

December 2024 - February 2025

Founding President & 1st President

February 2024 - November 2024

- Contributed to improving language proficiency and fostering academic development through reading and debating multilingual papers and books.
- Facilitated information exchange within the university and strengthened networking opportunities among student researchers.

Academic Service & Leadership**Teaching Assistant (TA)**

Physical Electronics (EEE3210), Spring 2026 February 2026 - June 2026 (Expected)
School of Electrical & Electronic Engineering, Yonsei University

Administrative Assistant

School of Electrical & Electronic Engineering, Yonsei University January 2026 - Present

Executive Member

Student Association March 2023 - February 2024
Major in Semiconductor Display, Gachon University